

Riley Strobe

Winter Quarter 2026

Introduction to Microprocessors

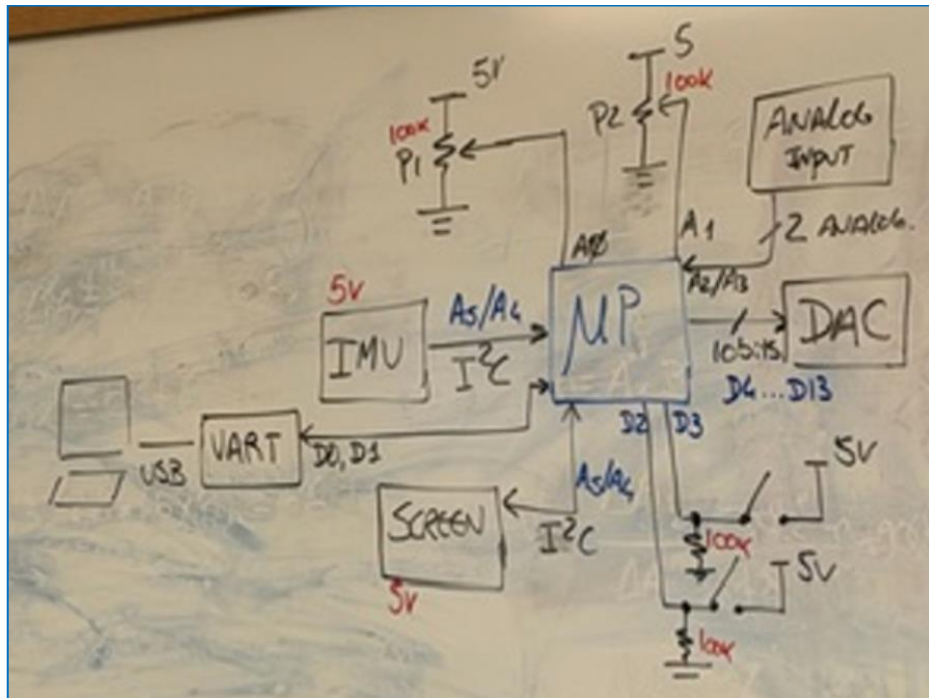
DSP Shield Design Documentation:

Introduction:

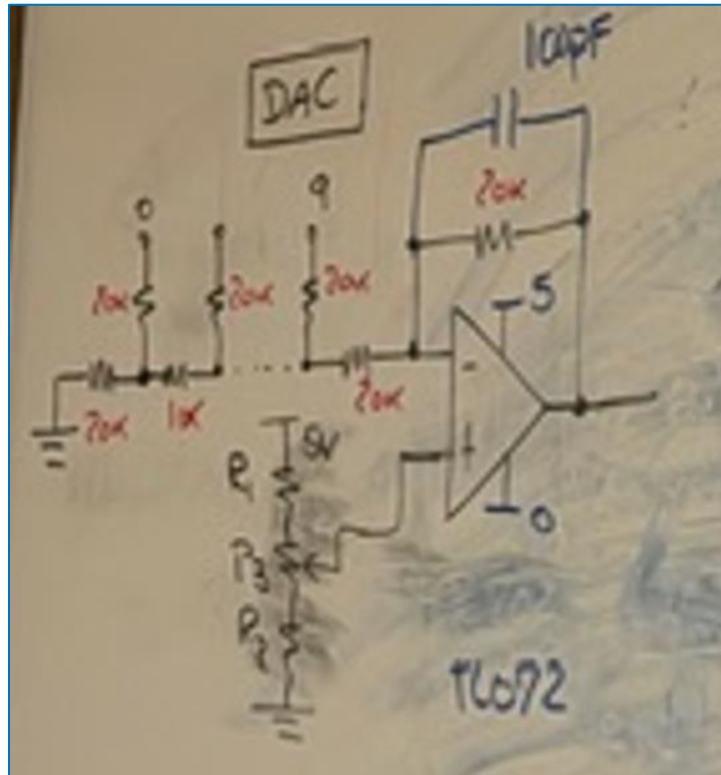
The objective of this project was to create a shield for an Arduino Uno R3. This shield will use an IMU, a screen, a DAC, two buttons, two analog input/output, and two potentiometers.

Design:

The shield will have the following block diagram:



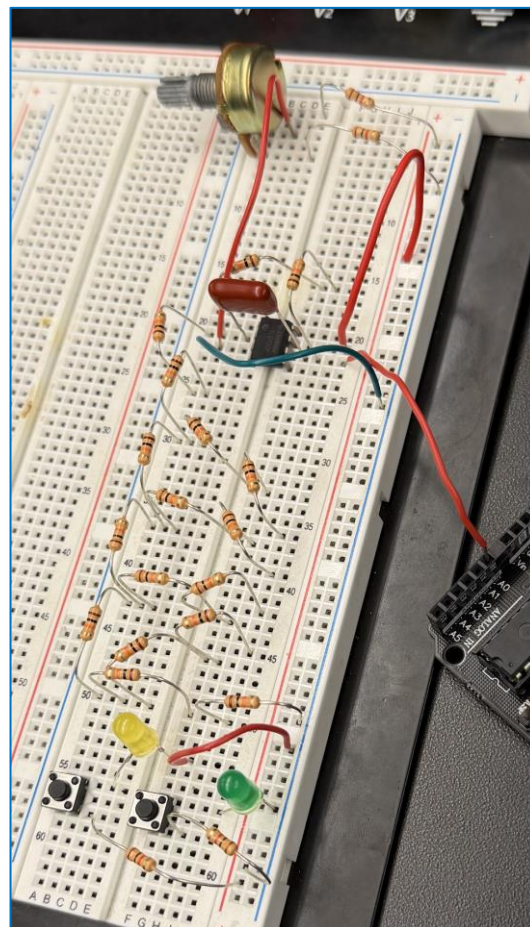
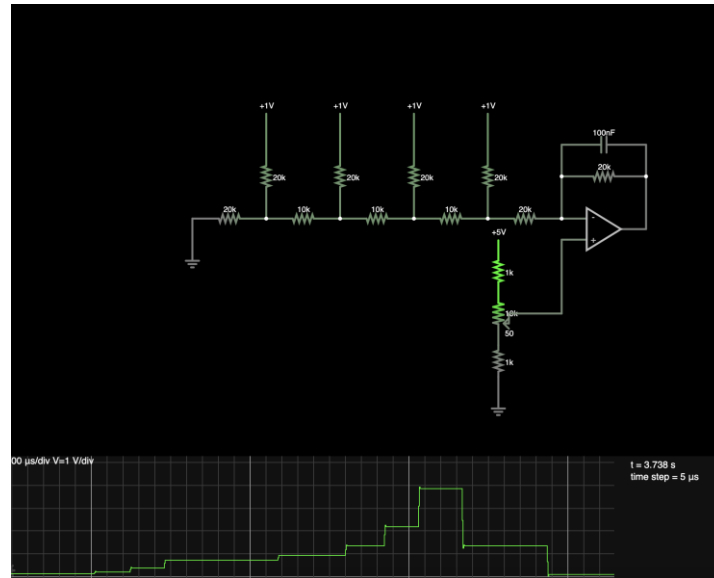
The 10-Bit-DAC had the following initial design:



During the assembly of the shield, the 20K and 10K resistors were replaced with 2K and 1K resistors respectively due to component availability.

DAC Prototyping:

Before purchasing the PCB board, the DAC design was tested. First, it was designed in Falstad. Then, the 4-Bit version was assembled on a protoboard and tested using the serial monitor while twisting the potentiometer.



```

DAC_4_Bit_Tester.ino

13   pinMode(bit1, OUTPUT);
14
15
16   }
17   void loop()
18   {
19
20
21     analogWrite(bit4, HIGH);
22     analogWrite(bit3, HIGH);
23     analogWrite(bit2, HIGH);
24     analogWrite(bit1, HIGH);
25
26     // read the output
27     gResult = analogRead(A0);
28
29     Serial.println(gResult);
30     delay(1000);
31   }

```

Output Serial Monitor X

Message (Enter to send message to 'Arduino UNO' on '/dev/cu.usbmodem1440')

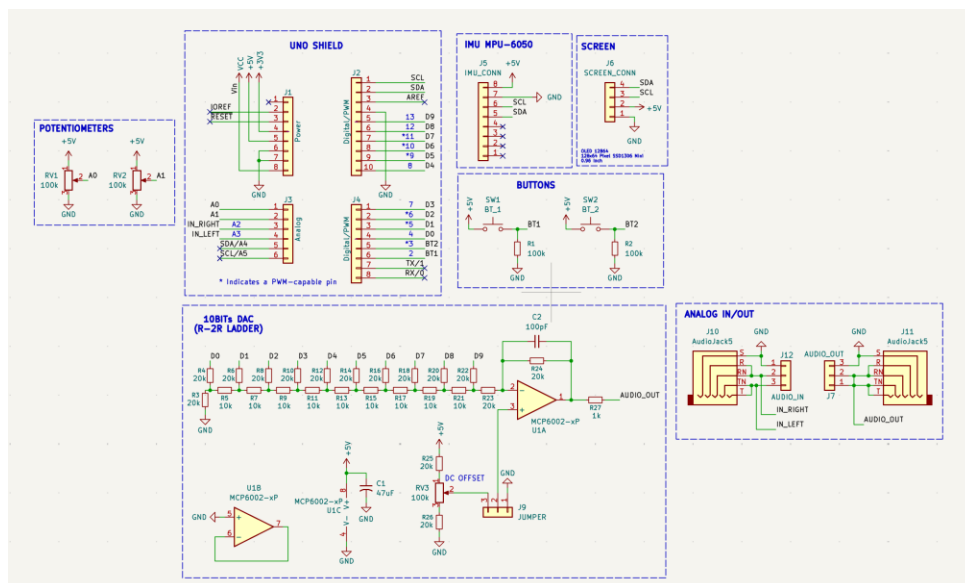
```

599
0
0
153
618
57
0
267

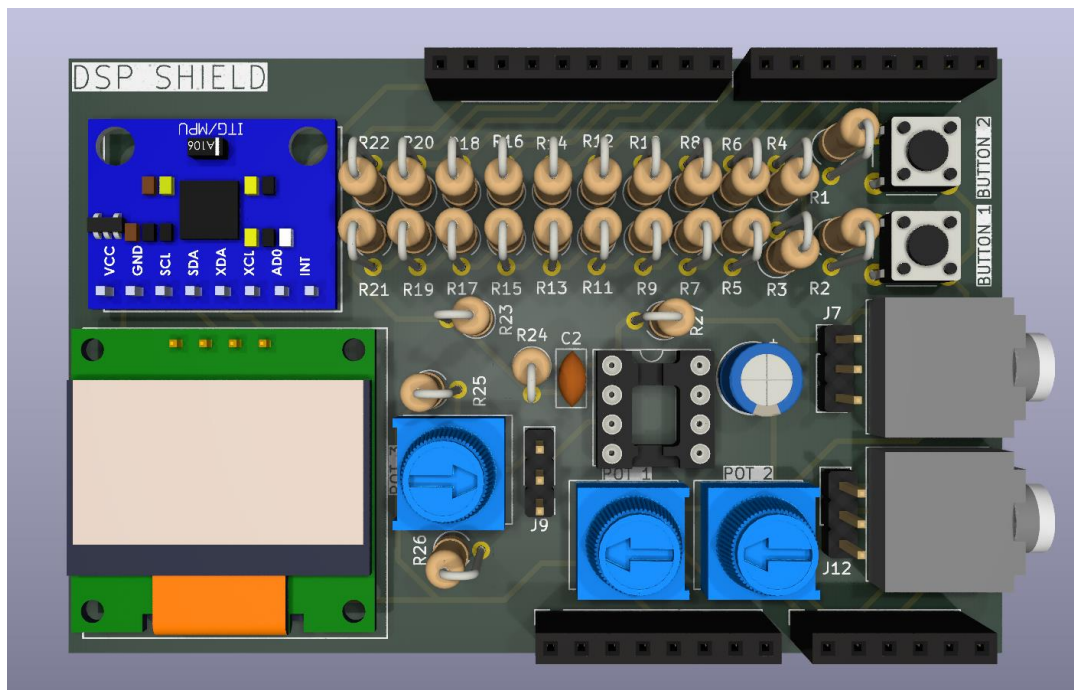
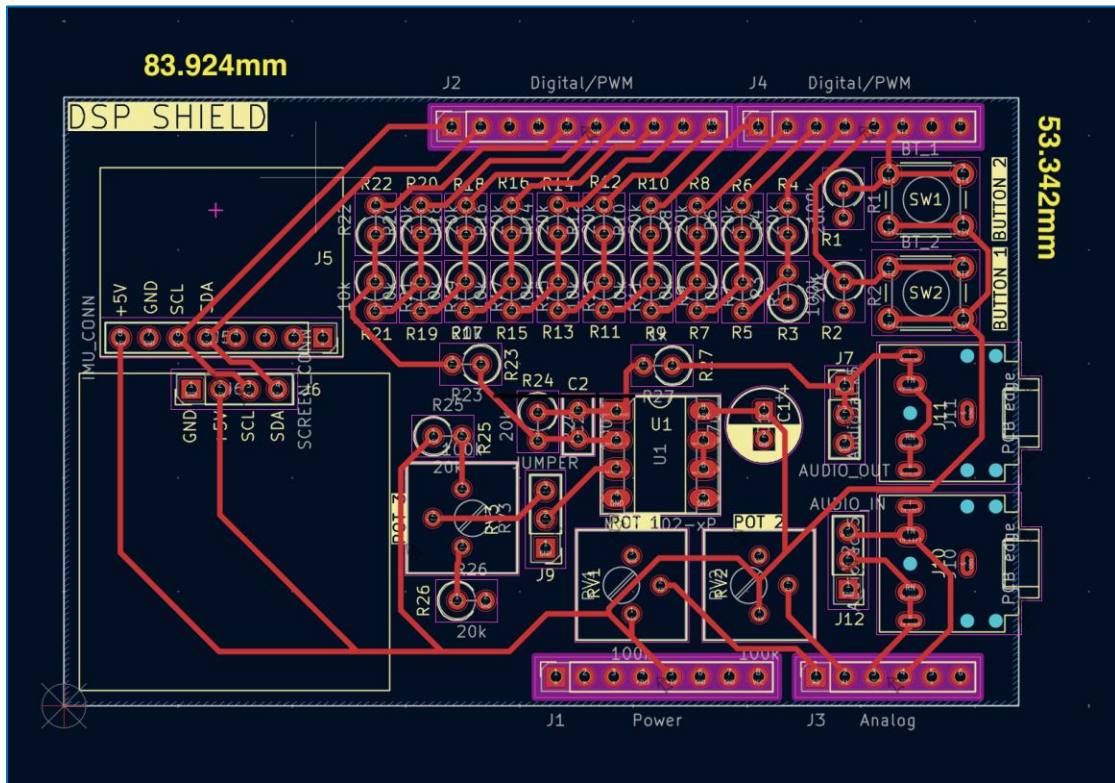
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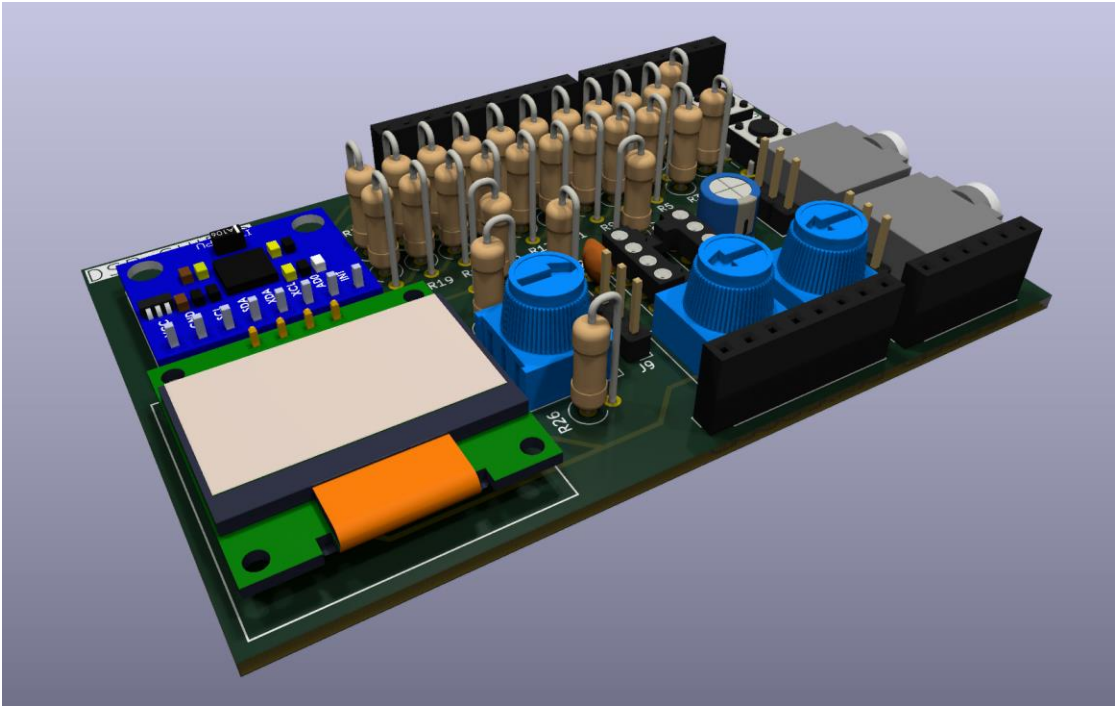
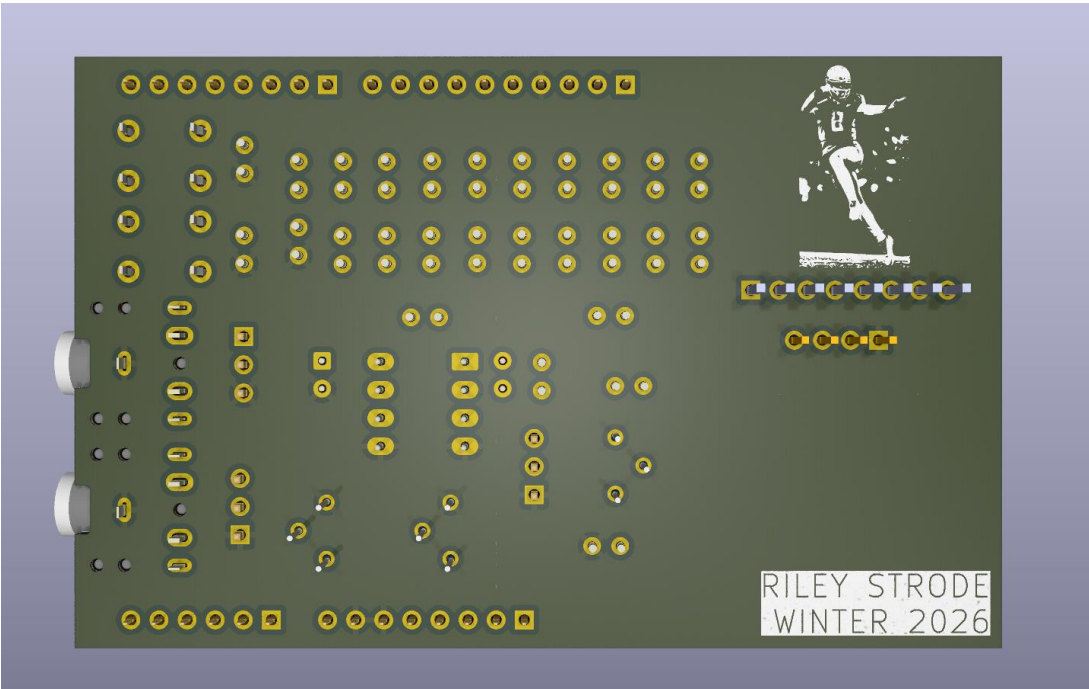
PCB Design:

Schematics



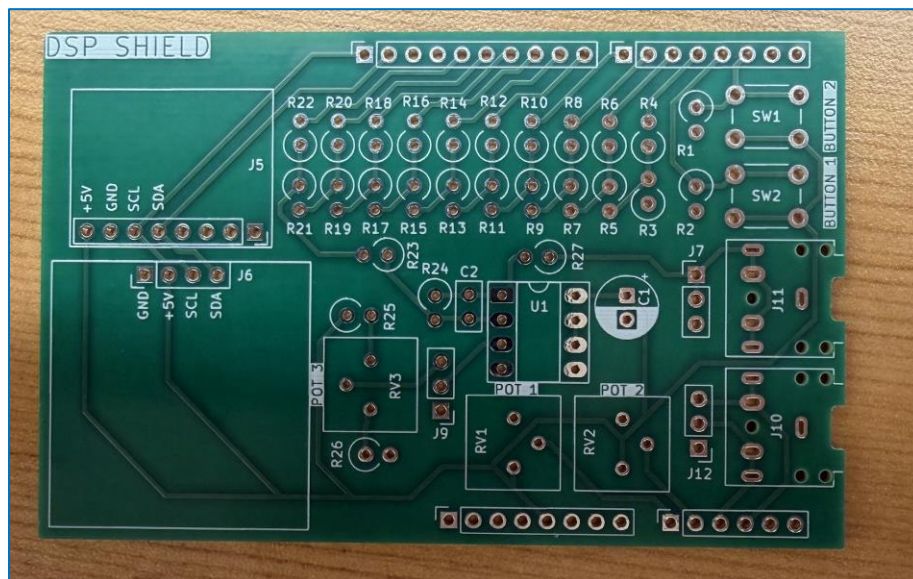
Board Layout

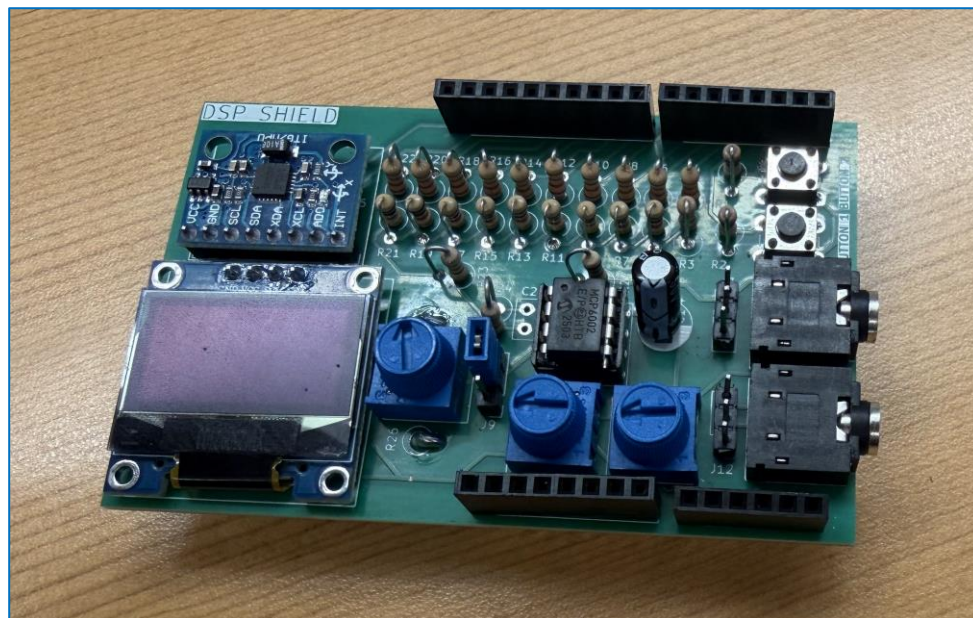
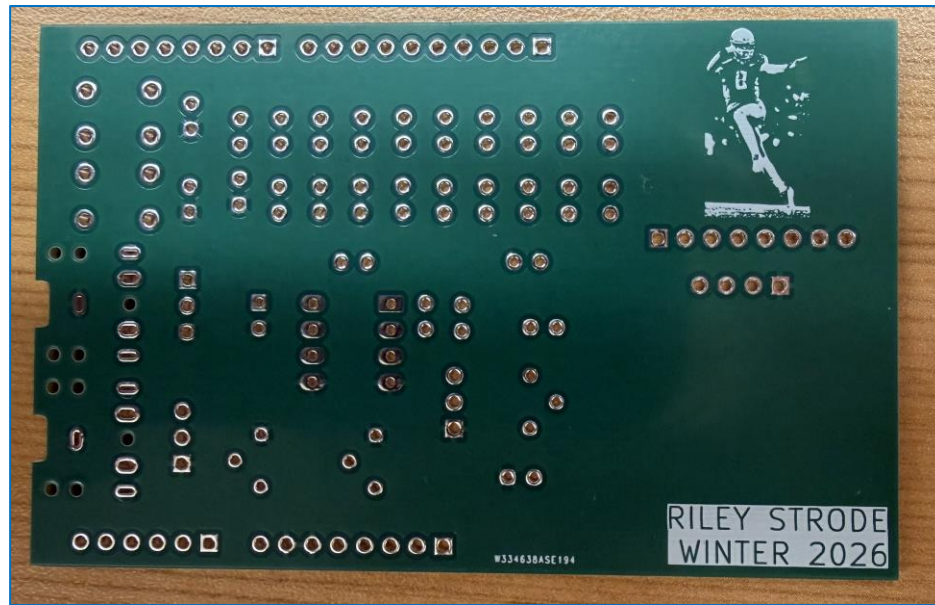


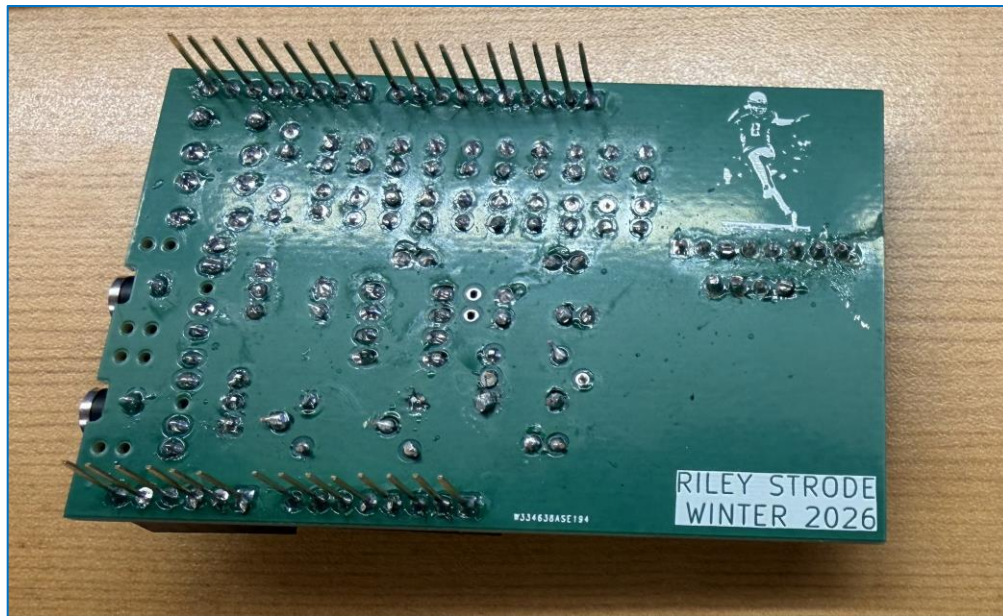


Assembly Stage:

After receiving my board, I soldered it by hand using lead free solder.







Testing:

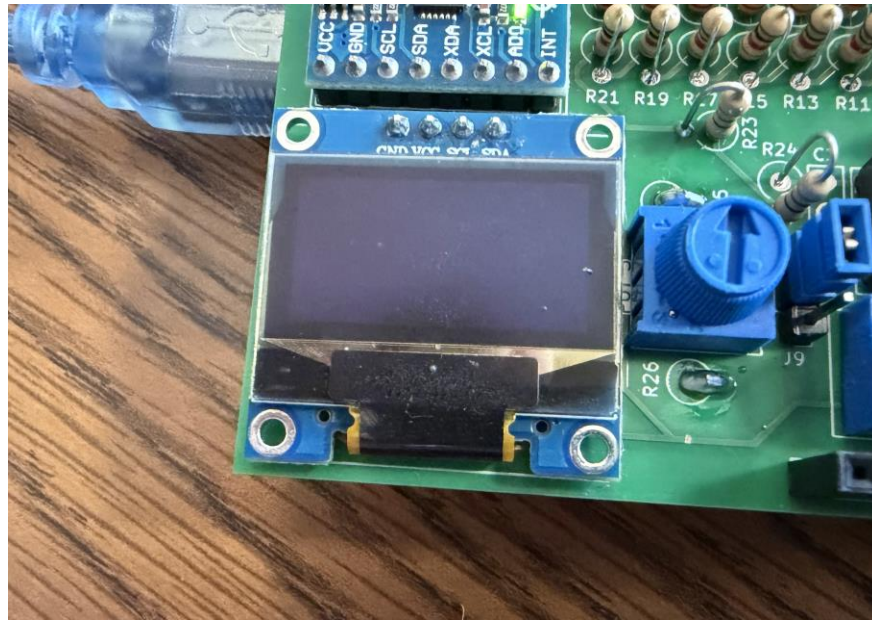
To check the functioning of the shield, the components needed to be tested.

Result from MPU Testing:

```
X-Accel = 0
X-Accel = 0
X-Accel = 0
X-Accel = 0
X-Accel = 0
X-Accel = 0
X-Accel = 0
```

This result shows that the MPU is not functioning correctly. As I move the shield, the X-Acceleration should be changing.

Results from Screen testing:



This result shows that the screen is not working correctly. It should be scrolling through 5 different images, but it is not even turning on. I'm guessing that the component was damaged during the soldering process.